

Path Testing

- The objective of path testing is to ensure that the set of test cases is such that each path through the program is executed at least once.
- The starting point for path testing is a program flow graph that shows nodes representing program decisions and arcs representing the flow of control.
- Statements with conditions are therefore nodes in the flow graph.

Steps for Basis Path testing

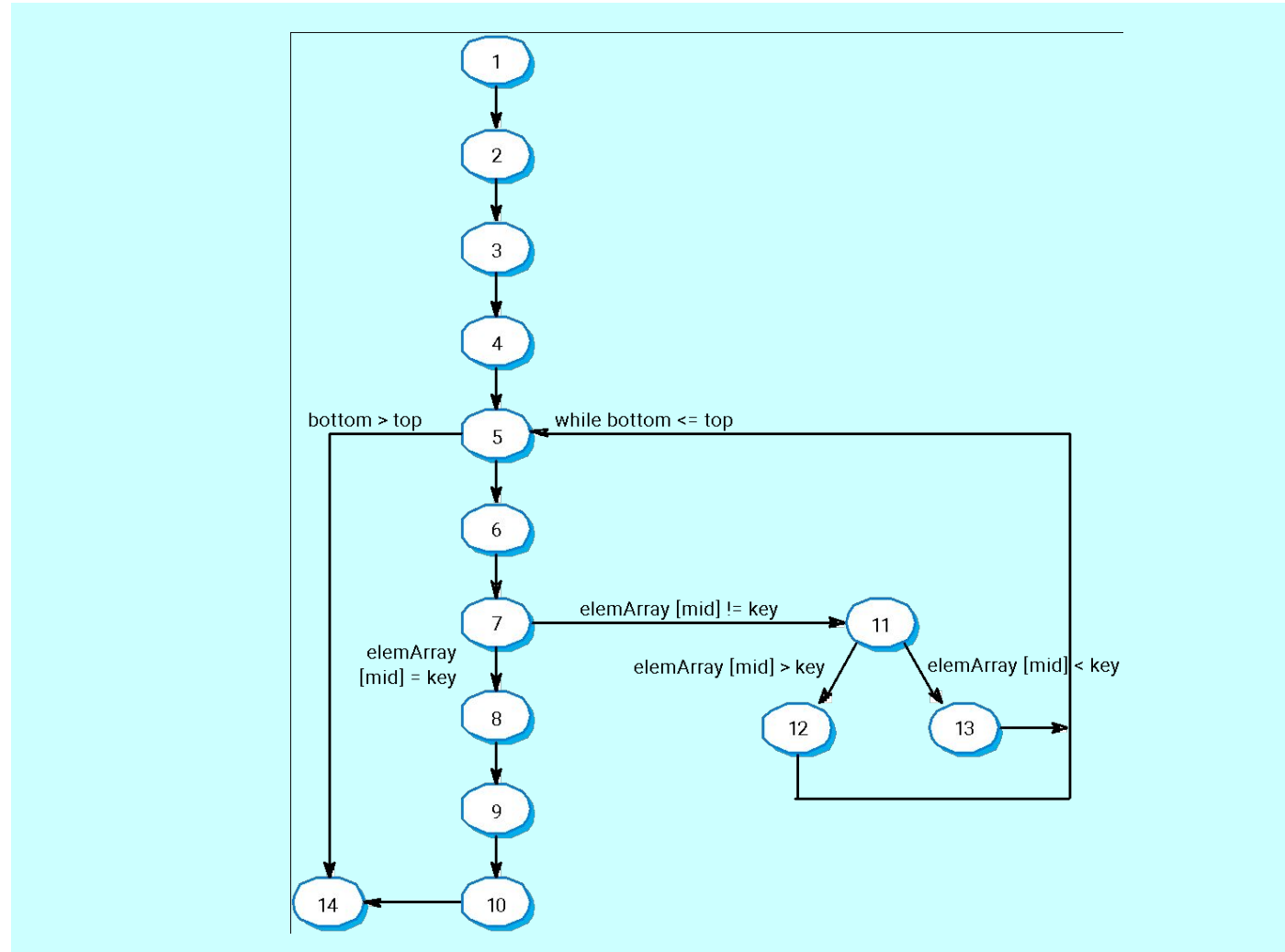
- The basic steps involved in basis path testing include
- Draw a control graph (to determine different program paths)
- Find a basis set of paths
- Generate test cases to exercise each path

```

public static void search ( int key, int 0elemArray, Result r )
{
1. int bottom =0 ;
2. int top =elemArray.length - 1 ; int mid;
3. r.found =false ;
4. r.index =-1 ;
5. while ( bottom <= top )
{
6 mid =(top + bottom) / 2 ;
7 if (elemArray [mid] = key)
{
8 rindex = mid;
9 r.found =true ;
10 return ; } // if part
else
{
11 if (elemArray [mid] < key)
12 bottom = mid + 1 ;
else
13 top = mid - 1 ; }
} //while loop
14. } //search

```

Binary Search Flow Graph



Independent Paths

- 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 14
- 1, 2, 3, 4, 5, 14
- 1, 2, 3, 4, 5, 6, 7, 11, 12, 5, ...
- 1, 2, 3, 4, 5, 6, 7, 11, 13, 5, ...
- Test cases should be derived so that all of these paths are executed
- A dynamic program analyser may be used to check that paths have been executed